

Uncertainty breakdown of the combined top-quark mass: three impact definitions

Math note for the two-stage combination (pyconvino cross-section combination $\rightarrow$ mtpole-ttj-pyconvino χ^2 mass fit). Covers (A) the classic freeze/conditional impact, (B) the marginal “global impact” of arXiv:2307.04007, and (C) the exact additive variance-share generalisation used for correlated priors.

All numbers quoted are **uncertainties only** (the fit is blinded); they come from the 3-experiment `Combination_ATLAS813CMS13_corrV2` fit (ATLAS 8 TeV + ATLAS 13 TeV + CMS 13 TeV, NNLO / CT18NNLO, `polyOrder 1`).

0. Setup and notation

Stage 1 — combination. `pyconvino` combines the input differential measurements into a vector of combined cross sections $x \in \mathbb{R}^n$ (here $n = 19$ bins), profiling N experimental nuisance parameters $\theta \in \mathbb{R}^N$ (here $N = 614$). Each nuisance carries a Gaussian prior

$$\theta \sim \mathcal{N}(0, C), \quad C^{-1} \equiv \text{inv_C (prior precision)}.$$

For the usual “unit, independent” priors $C = \mathbb{K}$; the `corrV2` setup deliberately makes C (hence `inv_C`) **non-diagonal**, correlating common systematics *across experiments*. The combination returns the joint post-fit covariance

$$\text{cov_full} = \begin{pmatrix} \Sigma_{xx} & \Sigma_{x\theta} \\ \Sigma_{\theta x} & \Sigma_{\theta\theta} \end{pmatrix}, \quad \Sigma_{xx} \equiv C_{\text{tot}} = \text{combined_covariance}, \quad \Sigma_{x\theta} = \text{x_sys_cov}.$$

Stage 2 — mass fit. In the Gaussian/linear regime the mass estimator is a linear functional of the combined cross sections,

$$\boxed{\hat{m} - m_0 = w^\top (x - x_0)} \quad w = \text{Fisher weights (_fisher_weights)}.$$

(The code applies a normalisation Jacobian J so that $w_{\text{abs}} = J^\top w$ acts in absolute-cross-section space; that is an implementation detail and is suppressed below.)

The one vector that drives everything. Define the covariance of the mass with each nuisance,

$$\boxed{g_i \equiv \text{Cov}(\hat{m}, \theta_i) = (w^\top \Sigma_{x\theta})_i} \quad (\text{code: } \text{dm_i} = w_{\text{abs}} \cdot \text{x_sys_cov}[:, i]).$$

Also write $\sigma_{\text{post},i} = \sqrt{\Sigma_{\theta\theta,ii}}$ (post-fit width of θ_i) and $\sigma_{\text{pre},i} = \sqrt{C_{ii}}$ (prior width). The total experimental mass variance is $\sigma_m^2 = w^\top C_{\text{tot}} w$.

Everything below is a different way of attributing σ_m^2 (or its systematic part) to the individual θ_i .

A. Classic impact — freeze / conditional method

Recipe. Fix one nuisance to its best fit $\pm$ its **post-fit** uncertainty, $\theta_i = \hat{\theta}_i \pm \sigma_{\text{post},i}$, re-minimise, read the shift in $\hat{m}$.

Closed form. For a Gaussian joint posterior of $(\hat{m}, \theta)$ the conditional mean gives

$$I_i^{\text{freeze}} = \frac{\text{Cov}(\hat{m}, \theta_i)}{\sigma_{\text{post},i}} = \frac{g_i}{\sigma_{\text{post},i}} = \sigma_m \rho(\hat{m}, \theta_i)$$

i.e. the familiar “impact = $\sigma_m \times$ correlation”.

Stat/syst split (Tier-1). The systematic is defined by *freezing all* nuisances:

$$\sigma_{\text{stat}}^{\text{freeze}} = (\text{mass unc. with all } \theta \text{ fixed}), \quad (\sigma_{\text{syst}}^{\text{freeze}})^2 = \sigma_m^2 - (\sigma_{\text{stat}}^{\text{freeze}})^2 = w^\top V_{\text{syst}} w,$$

with $V_{\text{syst}} = C_{\text{tot}} - V_{\text{stat}}$ (code: `total_syst_covariance`). Equivalently $f = (w^\top V_{\text{syst}} w) / (w^\top C_{\text{tot}} w)$ and $\sigma_{\text{syst}} = \sigma_m \sqrt{f}$, $\sigma_{\text{stat}} = \sigma_m \sqrt{1-f}$.

Key property / caveat. The freeze *stat/syst split closes* ($\sigma_{\text{stat}} \oplus \sigma_{\text{syst}} = \sigma_m$), **but the per-nuisance I_i^{freeze} do NOT add up to $\sigma_{\text{syst}}^{\text{freeze}}$** — the classic “impacts don’t sum” fact, because the θ_i are correlated in the post-fit and each I_i^{freeze} double-counts shared constraint.

Live numbers: $\sigma_{\text{stat}}^{\text{freeze}} = 424.7$ MeV, $\sigma_{\text{syst}}^{\text{freeze}} = 733.1$ MeV ($\oplus = 847.2$ MeV experimental total).

B. Global impact — marginal (arXiv:2307.04007)

Recipe. Shift the *external/auxiliary observable* y_i that constrains θ_i by its **prefit** width $\sigma_{\text{pre},i}$, let all θ **float**, refit, read $\Delta \hat{m}$.

Relation to the freeze impact. The two differ by the *post-fit nuisance-parameter covariance* factor $\sigma_{\text{post},i} / \sigma_{\text{pre},i}$:

$$I_i^{\text{glob}} = I_i^{\text{freeze}} \times \frac{\sigma_{\text{post},i}}{\sigma_{\text{pre},i}} = \frac{g_i}{\sigma_{\text{pre},i}} \xrightarrow{\text{unit priors } \sigma_{\text{pre}}=1} I_i^{\text{glob}} = g_i = \text{Cov}(\hat{m}, \theta_i).$$

This $I_i^{\text{glob}} = g_i$ is exactly the code’s marginal `dm_i` and the marginal forest bars.

Additive property (the reason the method exists). *When the auxiliary observables are independent* (C , hence `inv_C`, diagonal),

$$(\sigma_{\text{syst}}^{\text{glob}})^2 = \sum_i (I_i^{\text{glob}})^2 = \sum_i g_i^2.$$

Stat/syst interpretation (paper slide 8). The global split books the *in-situ statistical constraint of the nuisances* as **statistical**, so $\sigma_{\text{syst}}^{\text{glob}}$ = fluctuations of the external measurements only, and $\sigma_{\text{stat}}^{\text{glob}} = \sqrt{\sigma_m^2 - (\sigma_{\text{syst}}^{\text{glob}})^2}$ absorbs that in-situ constraint. Same total, different stat/syst line than the freeze method — neither is “more correct”.

Caveat under correlated priors. With `inv_C` non-diagonal (our `corrV2` case) the plain quadrature $\sum_i g_i^2$ is **not** σ_{syst}^2 — see § C.

Live numbers: $\sqrt{\sum_i g_i^2} = 954.7$ MeV, which even **exceeds** the experimental total (852.1 MeV) — a clear symptom that quadrature-summing the marginals is wrong here.

C. Global impact — exact additive variance share (correlated priors)

The correct systematic variance. Propagate fluctuations of the auxiliary data y (covariance = prior covariance $C = \text{inv_C}^{-1}$) through the profiled fit. Using $\partial\hat{x}/\partial y = \Sigma_{x\theta} \text{inv_C}$,

$$\frac{\partial\hat{m}}{\partial y} = w^\top \Sigma_{x\theta} \text{inv_C} = g^\top \text{inv_C},$$

$$\sigma_{\text{syst}}^2 = \frac{\partial\hat{m}}{\partial y} C \frac{\partial\hat{m}}{\partial y}^\top = (g^\top \text{inv_C}) \text{inv_C}^{-1} (\text{inv_C} g) = g^\top \text{inv_C} g$$

This is positive ($\text{inv_C} \succ 0$) and provably $\leq \sigma_m^2$; it reduces to $\sum_i g_i^2$ only when $\text{inv_C} = \mathbb{K}$.

Exact additive attribution. Define $a \equiv \text{inv_C} g$. Then

$$\sigma_{\text{syst}}^2 = g^\top a = \sum_i \underbrace{g_i a_i}_{s_i}, \quad s_g = \sum_{i \in g} g_i a_i, \quad \boxed{\sum_i s_i = \sum_g s_g = \sigma_{\text{syst}}^2 \text{ (exactly)}}$$

Plot the signed root $\text{sign}(s)\sqrt{|s|}$ so the (signed) squares still add to σ_{syst}^2 .

Why this is the paper's method, generalised. With independent priors $\text{inv_C} = \mathbb{K} \Rightarrow a_i = g_i \Rightarrow s_i = g_i^2 = (I_i^{\text{glob}})^2$ — the two agree term by term. Restoring the correlation metric inv_C is the *only* change; it turns the oblique (correlated) axes into the right inner product.

Geometric picture — why the marginals (§ B) don't sum but these do. σ_{syst} is the *length* of g in the inv_C metric. Cholesky $\text{inv_C} = LL^\top$ and $h = L^\top g$ give an **orthogonal** (whitened) basis where Pythagoras holds exactly, $\sum_k h_k^2 = g^\top \text{inv_C} g$ (verified to 10^{-13}). The named nuisances are the *oblique* axes; summing the squares of their projections g_i^2 ignores the cross terms $\sum_{i \neq j} g_i (\text{inv_C})_{ij} g_j$. The variance share $s_i = g_i a_i$ folds each row's full cross term back in, restoring closure.

Why a share can be negative. $s_i < 0 \iff \text{sign}(g_i) \neq \text{sign}(a_i)$: the source's own mass pull g_i opposes the correlation-weighted pull $a_i = (\text{inv_C} g)_i$, i.e. it is *anti-correlated through the prior* with the dominant systematics, so including it slightly **reduces** the total variance — the exact analogue of a negative covariance term in an ordinary correlated error budget.

Live numbers: $\sqrt{g^\top \text{inv_C} g} = 830.3 \text{ MeV (linear)} \leq 852.1 \text{ total}$; per-group shares close to $\sigma_{\text{syst}} = 825.5 \text{ MeV (HESSE-anchored)}$ with **1 of 49** groups negative (**ATLAS_13TeV_modelling_singleTOP**, -1.9 MeV : $g = -1.7$, $a = +2.2$); per-nuisance shares close to the same, 23 of 614 negative.

D. Side-by-side

	A. Freeze / classic	B. Global marginal	— C. Global — additive share
Per-source formula	$I_i^{\text{freeze}} = g_i / \sigma_{\text{post},i}$	$I_i^{\text{glob}} = g_i / \sigma_{\text{pre},i}$	$s_i = g_i (\text{inv_C } g)_i$
Reads as	“shift if θ_i fixed to $\pm\sigma_{\text{post}}$ ”	“shift if aux obs moves $\pm\sigma_{\text{pre}}$ ”	“contribution of θ_i to σ_{syst}^2 ”
$\sum$ over sources = σ_{syst}^2 ?	✗ (never)	✓ only if priors independent	✓ always (exact)
Sign	≥ 0	≥ 0	signed (can be < 0)
Books in-situ NP constraint as	systematic	statistical	(attribution of the global σ_{syst})
Total exp. syst	733.1 MeV	(marginals overshoot: 954.7)	825.5 MeV
Code output	doExpStatSyst Breakdown (Tier-1)	global_impacts_ {groups,nuisances}	global_impacts_{groups, nuisances}_additive

Relations: $I_i^{\text{glob}} = I_i^{\text{freeze}} \cdot \sigma_{\text{post},i} / \sigma_{\text{pre},i}$; and $s_i \xrightarrow{\text{inv_C} \rightarrow \mathbb{K}} (I_i^{\text{glob}})^2$. Both stat/syst splits close to the same experimental total (847.2 MeV here); they only draw the stat/syst line in different places.

E. Practical guidance

- **Headline stat vs syst:** either split is defensible; quote which convention (freeze vs global) is used. Freeze = traditional; global = “external-only systematic”.
- **“Which sources dominate?”** — read the **marginal** bars (§ B): honest “size of each source on its own”. Do **not** quadrature-sum them under correlated priors.
- **“Budget that adds up to σ_{syst} ”** — use the **additive shares** (§ C). Expect a few small negative entries (anti-correlated sources); that is correct.
- Everything here is the *experimental* systematic (in the combined-cross-section covariance). Theory (PDF, scale, interpolation) is added separately: $\sigma_{\text{total}}^2 = \sigma_{\text{exp}}^2 + \sigma_{\text{PDF}}^2 + \sigma_{\text{interp}}^2 + \sigma_{\text{scale}}^2$.

Reference: A. Freitas et al., “*Uncertainty breakdown via global impacts*”, arXiv:2307.04007 (CMS-week slides by A. Gilbert, slides 4–9). § A $\equiv$ conditional method (slide 3); § B $\equiv$ global impact / marginal additive form (slides 5–6, independent-aux case); § C $\equiv$ its exact quadratic-form generalisation for correlated auxiliary observables.